

## SCALAR CLASSES IN CRITICAL HALL MULTIPLICATION

### 1.1 THE LOWEST COEFFICIENT AT A SCALAR TRIPLE

At a scalar triple, every subspace is invariant. Its Hall coefficient still depends on the ambient cubic potential. Complete flags produce a distinguished class in this coefficient, including at the scalar point. The class determines the lowest-degree images for the potentials of ranks three, four, and five.

All spaces are complex algebraic varieties with their analytic topology. Coefficients are rational. For  $1 \leq n \leq 5$ , put

$$V_n = \text{Mat}_n(\mathbb{C})^3, \quad f_n(A, B, C) = \text{Tr}(A[B, C]), \quad Z_n = \{[A, B] = [B, C] = [C, A] = 0\}.$$

Differentiation using the trace pairing gives  $Z_n = \text{Crit}(f_n)$ . Write  $\mathfrak{X}_n = [V_n/\text{GL}_n]$  and  $\mathfrak{M}_n = [Z_n/\text{GL}_n]$ . Constructible complexes on these quotient stacks retain the conjugation action and its descent data. Their ordinary atlas pullbacks have no additional shift. The coefficient is

$$\mathcal{P}_n = \Phi_{f_n} \mathbb{Q}_{\mathfrak{X}_n}[2n^2] \big|_{\mathfrak{M}_n}, \quad \Phi_f K = \text{Cone}(i^* K \longrightarrow \psi_f K)[-1],$$

where  $i : f^{-1}(0) \hookrightarrow V$  and  $\psi_f$  denotes nearby cycles. This normalization agrees with Massey[1, pp. 1–2]. Thus  $H^j(K[a]) = H^{j+a}(K)$  and  $\Phi_0 K = K$ . The tensor differential is  $d(x \otimes y) = dx \otimes y + (-1)^{|x|} x \otimes dy$ . No completion or change of coefficients is used.

For  $s = (aI_n, bI_n, cI_n)$ , write  $P_{n,s}$  for the ordinary atlas stalk of  $\mathcal{P}_n$ . Translation by scalar matrices preserves  $f_n$ , so it suffices to compute  $P_{n,0}$ . Define the commuting-pair variety and its complement by

$$C_n = \{(A, C) \in \text{Mat}_n(\mathbb{C})^2 : [C, A] = 0\}, \quad U_n = \text{Mat}_n(\mathbb{C})^2 \setminus C_n.$$

Borel–Moore homology below uses locally finite rational chains and the complex orientation of each smooth stratum.

**PROPOSITION 1.1.1** (*Scalar stalk and commuting pairs*). For  $2 \leq n \leq 5$  and every integer  $j$ , there are isomorphisms

$$H^j(P_{n,s}) \simeq \tilde{H}^{j+2n^2-1}(U_n; \mathbb{Q}) \simeq H_{2n^2-j}^{\text{BM}}(C_n; \mathbb{Q}). \quad (1.1)$$

In particular,

$$H^j(P_{n,s}) = 0 \quad (j < -2n), \quad H^{-2n}(P_{n,s}) \simeq \mathbb{Q}.$$

The same last two assertions hold for  $n = 1$ , where  $P_{1,s} = \mathbb{Q}[2]$ . Potential monodromy acts as the identity on the entire scalar stalk complex for every rank under consideration.

*Proof.* Use the positive real ray in the potential plane. By Massey[1, §1, p. 2], the normalized vanishing-cycle stalk is computed by cochains with support in  $P = \{\text{Re } f_n \leq 0\}$ , restricted to the origin. The fibre

convention is the relative-cochain convention  $\text{Fib}(C^*(B) \rightarrow C^*(B \setminus P))$ . Both  $P$  and its complement are invariant under positive real scaling. The radial map

$$x \mapsto \frac{x}{1 + \|x\|/\delta}$$

maps  $V_n$  into the ball of radius  $\delta$ . Interpolation of the denominator gives homotopy inverses for the inclusions of these pairs. Thus their relative cohomology computes the scalar stalk as  $\delta$  decreases to zero.

Let  $F_n = f_n^{-1}(1)$  and  $H = \{z \in \mathbb{C} : \text{Re } z > 0\}$ . Choose the cubic root on  $H$  that is positive on the positive real axis. Homogeneity gives mutually inverse homeomorphisms

$$V_n \setminus P \longrightarrow F_n \times H, \quad x \mapsto (f_n(x)^{-1/3}x, f_n(x)), \quad (y, z) \mapsto z^{1/3}y.$$

Since  $V_n$  and  $H$  are contractible, the support complex has cohomology  $\tilde{H}^{k-1}(F_n; \mathbb{Q})$  in degree  $k$ . This argument fixes the local-to-global comparison at the scalar point.

The cyclic trace identity writes the actual potential as  $f_n = \text{Tr}(B[C, A])$ . Projection onto  $(A, C)$  gives

$$F_n \longrightarrow U_n, \quad (A, B, C) \mapsto (A, C),$$

an affine bundle of complex rank  $n^2 - 1$ . Indeed, its fibre is the affine hyperplane  $\text{Tr}(BD) = 1$ , where  $D = [C, A] \neq 0$ . The trace pairing is nondegenerate. A continuous section is

$$B_0(A, C) = \frac{D^*}{\text{Tr}(D^*D)},$$

where  $D^*$  is the conjugate transpose. The affine homotopy from  $B$  to  $B_0$  stays in this hyperplane. Therefore projection induces the first isomorphism in (1.1), after the shift  $[2n^2]$ . Projection is conjugation equivariant. The chosen section only supplies an ordinary topological homotopy inverse.

The real dimension of the pair space is  $4n^2$ . Its only nonzero Borel–Moore group is  $H_{4n^2}^{\text{BM}} = \mathbb{Q}$ . The closed-open exact sequence for  $C_n$  and  $U_n$ , followed by Poincaré duality on the smooth open set  $U_n$ , gives

$$\tilde{H}^k(U_n; \mathbb{Q}) \simeq H_{4n^2-k-1}^{\text{BM}}(C_n; \mathbb{Q}).$$

For  $k = 0$ , the reduced group removes the ambient fundamental class. The complement of a proper complex subvariety in this affine space is connected: a generic complex line joins each point to a common generic intermediate point while avoiding finitely many intersections. The remaining nonpositive degrees vanish. Substitution gives the second isomorphism in (1.1).

For completeness, the top dimension of  $C_n$  follows directly from Jordan forms. Fix a Jordan type for  $A$  with  $r$  distinct eigenvalues and a centralizer of dimension  $c$ . Its matrix stratum has dimension  $n^2 - c + r$ . The matrices  $C$  commuting with  $A$  form a vector bundle of rank  $c$  on this stratum. The corresponding pair stratum has dimension  $n^2 + r$ . There are finitely many Jordan types. The unique stratum with  $r = n$  has dimension  $n^2 + n$  and is irreducible: ordered eigenvalues and diagonal entries of  $C$  parameterize it, together with a change of basis. The finite permutation quotient preserves irreducibility. Every other stratum has dimension at most  $n^2 + n - 1$ . Consequently the top Borel–Moore group is

$$H_{2n^2+2n}^{\text{BM}}(C_n; \mathbb{Q}) = \mathbb{Q}, \quad H_m^{\text{BM}}(C_n; \mathbb{Q}) = 0 \quad (m > 2n^2 + 2n).$$

The top generator is the complex fundamental class of the closure of that stratum. No assertion about lower-dimensional components is needed. Equation (1.1) gives the degree bounds.

Finally, at the scalar origin the maps

$$(A, B, C) \mapsto (A, e^{i\theta} B, C), \quad 0 \leq \theta \leq 2\pi,$$

preserve every centered Euclidean ball and multiply  $f_n$  by  $e^{i\theta}$ . They trivialize the Milnor fibres over each circle in the potential plane. At  $2\pi$  the map is the identity. These trivializations commute with restriction to smaller balls, with the constant-to-nearby-cycle map, and with conjugation. They give identity potential monodromy on the scalar fibre complex. Translation gives the same assertion at every scalar triple.  $\square$

The first nonzero groups for the three target potentials are therefore

| $n$ | $\dim_{\mathbb{C}} C_n$ | first reduced Milnor degree | first degree of $P_{n,s}$ |
|-----|-------------------------|-----------------------------|---------------------------|
| 3   | 12                      | 11                          | -6                        |
| 4   | 20                      | 23                          | -8                        |
| 5   | 30                      | 39                          | -10.                      |

Each displayed group has rank one. The calculation concerns ordinary atlas stalks. It does not replace equivariant coefficient complexes by these graded vector spaces.

The rank-two scalar fibre gives a further check on the shifts. Write the traceless parts of  $A, B, C$  in the basis  $\text{diag}(1, -1), e_{12}, e_{21}$  of  $\mathfrak{sl}_2$ . If  $T$  is the matrix of their three coordinate columns, direct expansion gives  $f_2 = 2 \det T$ . The scalar directions are free. The determinant-one fibre retracts onto  $\text{SU}(3)$  by polar decomposition. The fibration  $\text{SU}(2) \rightarrow \text{SU}(3) \rightarrow S^5$  has fibre  $S^3$ . Its rational Serre spectral sequence has nonzero terms only in bidegrees  $(0, 0), (0, 3), (5, 0), (5, 3)$ , so every differential vanishes. The reduced fibre cohomology has degrees 3, 5, 8, each of rank one. After the normalized shift [8], the scalar coefficient has degrees -4, -2, 1.

## 1.2 COMPLETE FLAGS DEFINE SCALAR GENERATORS

The rank-one coefficient is  $\mathbb{Q}[2]$ . Complete flags let these coefficients enter the higher-rank critical coefficient without a choice of a scalar Milnor cycle. Let  $\text{Fl}_n$  be the variety of complete flags in  $\mathbb{C}^n$  and set  $p_n = n(n-1)/2$ . Let  $E_n^{\text{fl}} \subset V_n \times \text{Fl}_n$  classify triples preserving a complete flag. Its matrices are upper triangular in an adapted basis, so  $f_n|_{E_n^{\text{fl}}} = 0$ . The normal bundle of this incidence has complex rank  $3p_n$ . Its positive complex Thom class followed by flag integration defines

$$\gamma_n^{\text{fl}} : Rq_{n*} \mathbb{Q}_{E_n^{\text{fl}}} \longrightarrow \mathbb{Q}_{V_n}[4p_n],$$

where  $q_n$  forgets the flag. The map is proper because  $\text{Fl}_n$  is projective.

Pass to quotient stacks, and retain the full critical fibre product

$$\mathcal{T}_n = [E_n^{\text{fl}}/\text{GL}_n] \times_{\mathfrak{X}_n} \mathfrak{M}_n, \quad b_n : \mathcal{T}_n \longrightarrow \mathfrak{M}_n.$$

Thus the off-diagonal commutator equations remain part of the source. Proper exchange for vanishing cycles [2, §3, p. 35] and  $\Phi_0 = 1$  give

$$\nu_n : Rb_{n*} \mathbb{Q}_{\mathcal{T}_n}[2n] \longrightarrow \mathcal{P}_n, \quad 2n + 4p_n = 2n^2.$$

The unit of pullback and pushforward then defines degree-zero morphisms

$$h_n : \mathbb{Q}_{\mathfrak{M}_n}[2n] \longrightarrow Rb_{n*} \mathbb{Q}_{\mathcal{T}_n}[2n] \xrightarrow{\nu_n} \mathcal{P}_n, \quad \sigma_n = \frac{h_n}{n!}. \quad (1.2)$$

Every construction commutes with changes of frame. Its normal bundles and flag cycles retain their complex orientations on the action diagram. Thus (1.2) defines morphisms on the quotient stacks.

The sign of  $h_n$  can be determined at a matrix  $A$  with distinct eigenvalues  $\lambda_1, \dots, \lambda_n$ . In a transverse eigenbasis chart, the actual summand for  $i < j$  is

$$(\lambda_i - \lambda_j)(b_{ij}c_{ji} - b_{ji}c_{ij}).$$

Set  $\delta_{ij} = \lambda_i - \lambda_j \neq 0$  and introduce

$$\widehat{c}_{ij} = \delta_{ij}c_{ij}, \quad \widehat{c}_{ji} = \delta_{ij}c_{ji}, \quad q = b_{ij}\widehat{c}_{ji} - b_{ji}\widehat{c}_{ij}.$$

Each rescaling is complex linear with real determinant  $|\delta_{ij}|^2 > 0$ . Both together have determinant  $|\delta_{ij}|^4 > 0$ , so they preserve the complex normal orientation. The  $B$ -polarized positive disk has normalized coordinates

$$(b_{ij}, b_{ji}, \widehat{c}_{ij}, \widehat{c}_{ji}) = (u, v, -\bar{v}, \bar{u}), \quad q = |u|^2 + |v|^2.$$

Its complex parameter orientation defines a positive support generator. Projection to the normalized lower normal coordinates is  $(u, v) \mapsto (v, \bar{u})$ , of real degree  $-1$ . In the original coordinates it is  $(u, v) \mapsto (v, \delta_{ij}^{-1}\bar{u})$ . The additional complex multiplication has positive real determinant  $|\delta_{ij}|^{-2}$ , so the projection degree remains  $-1$ . The lower  $A$  normal cancels the flag tangent by multiplication with the nonzero eigenvalue difference. This complex linear map has positive real determinant. If  $u_n$  denotes the product of the  $B$ -polarized generators, put

$$e_n = (-1)^{p_n} u_n.$$

All support Thom degrees are even. Their ordered external products introduce no sign. Thus each complete flag contributes  $e_n$ , and there are  $n!$  complete flags. This proves

$$h_n|_{\text{distinct}} = n!e_n, \quad \sigma_n|_{\text{distinct}} = e_n. \quad (1.3)$$

Here  $\mathcal{P}_n = \mathbb{Q}[2n]$  on this locus, since its normal quadratic has  $2(n^2 - n)$  variables.

**THEOREM 1.2.1** (*Extension of the normalized lowest coefficient*). For  $1 \leq n \leq 5$ , the stalk of  $\sigma_n$  at any scalar triple induces an isomorphism

$$\mathbb{Q} \xrightarrow{\sim} H^{-2n}(\mathcal{P}_{n,s}).$$

Denote the image of 1 by  $v_{n,s}$ . On the scalar locus, these classes form a constant, conjugation-equivariant cohomology sheaf. They have identity potential monodromy.

*Proof.* Positive scaling of all three matrices preserves  $Z_n$  and each flag incidence, and multiplies  $f_n$  by a positive cube. It preserves the support half-plane, the flag cycles and the Thom orientations. Hence the section of  $\mathcal{H}^{-2n}(\mathcal{P}_n)$  induced by  $h_n$  is equivariant under positive scaling.

Suppose this section had zero stalk at the origin on the atlas. By the definition of a sheaf stalk, it would vanish on a neighborhood of the origin. Every point of  $Z_n$  can be scaled into that neighborhood. Scaling equivariance would force the section to vanish everywhere. This contradicts (1.3). Therefore  $h_n$  has a nonzero scalar stalk in degree  $-2n$ . Proposition 1.1.1 shows that this stalk group is one-dimensional. Division by  $n!$  gives the stated isomorphism.

Scalar translations preserve the function, incidences and orientations. They transport this generator to every scalar triple. The morphism already descends under conjugation, so its lowest cohomology sheaf has the trivial scalar stabilizer action. The monodromy assertion also follows from Proposition 1.1.1.  $\square$

This theorem specifies the lowest cohomology sheaf. It supplies no splitting of the higher cohomology or of the equivariant derived object.

### 1.3 THE SCALAR IMAGES OF THE HALL COEFFICIENT MAPS

Fix positive integers  $d, e$  with  $n = d + e \leq 5$ . The product order is submodule first, followed by quotient. Let  $E_{d,e}$  classify triples preserving a  $d$ -plane, and write

$$\mathcal{F}_{d,e} = [E_{d,e}/\mathrm{GL}_n] \times_{\mathfrak{X}_n} \mathfrak{M}_n, \quad \mathfrak{M}_d \times \mathfrak{M}_e \xleftarrow{a_{d,e}} \mathcal{F}_{d,e} \xrightarrow{b_{d,e}} \mathfrak{M}_n.$$

The diagonal block potential is  $f_d + f_e$ . The normal Thom map has degree  $6de$ , and Grassmannian integration has degree  $-2de$ . Their composite is

$$\gamma_{d,e} : Rq_{d,e*} \mathbb{Q}_{E_{d,e}} \longrightarrow \mathbb{Q}_{V_n} [4de].$$

For analytic functions  $f, g$ , the relative-cochain product and the support inclusion

$$\{\mathrm{Re} f \leq 0\} \times \{\mathrm{Re} g \leq 0\} \subset \{\mathrm{Re}(f + g) \leq 0\}$$

define the morphism  $\Phi_f K \boxtimes \Phi_g L \rightarrow k^* \Phi_{f+g}(K \boxtimes L)$ . The product and support inclusion are those of Massey[1, Lemma 1.2, p. 3, and Proposition 1.4, p. 5]. Here  $k$  includes the product of zero fibres in the zero fibre of the sum. This morphism is natural in coefficients and proper direct image: both assertions follow from the same maps of relative cochain complexes. For proper direct image, apply proper base change and the proper Künneth isomorphism[2, §1, pp. 13–14]. Its ordered triple products agree because they use the same product cochains and the same inclusion into the support of  $f + g + h$ .

Use this support morphism, smooth ambient block pullback, proper exchange, and  $\Phi(\gamma_{d,e})$ . Restriction to the full critical fibre product defines the actual coefficient map

$$\mu_{d,e} : Rb_{d,e*} a_{d,e}^* (\mathcal{P}_d \boxtimes \mathcal{P}_e) \longrightarrow \mathcal{P}_n. \quad (1.4)$$

Its shift identity is  $2d^2 + 2e^2 + 4de = 2n^2$ . The critical block map need not be smooth. The construction uses the support comparison as a morphism, including for  $f_2 + f_3$ . Its invertibility is unnecessary here.

Let  $\eta_{d,e}$  denote the unit  $\mathbb{Q}_{\mathfrak{M}_n} \rightarrow Rb_{d,e*} \mathbb{Q}_{\mathcal{F}_{d,e}}$ . The canonical tensor identification and the maps  $\sigma_d, \sigma_e$  give a morphism from  $\mathbb{Q}_{\mathcal{F}_{d,e}} [2n]$  to the incoming coefficient in (1.4).

**THEOREM 1.3.1** (*Scalar lowest-degree Hall images*). For  $d + e = n \leq 5$ , the following equality holds as morphisms of constructible complexes on  $\mathfrak{M}_n$ :

$$\mu_{d,e} \circ Rb_{d,e*} a_{d,e}^* (\sigma_d \boxtimes \sigma_e) \circ \eta_{d,e} [2n] = \binom{n}{d} \sigma_n. \quad (1.5)$$

At a scalar triple, the source and target of the induced map in degree  $-2n$  are both one-dimensional. In the generators specified by  $v_{d,s}, v_{e,s}$  and  $v_{n,s}$ , this map is multiplication by  $\binom{n}{d}$ . Its image is the whole lowest-degree target and its kernel in that degree is zero.

*Proof.* Refine an invariant  $d$ -plane by a complete flag in that plane and a complete flag in its quotient. The result is exactly a complete flag in  $\mathbb{C}^n$ . This identifies the refinement square on the full critical stacks, including its frame changes. Proper base change identifies the corresponding iterated direct images.

The forbidden lower entries for the refined flag are the disjoint union of internal entries in the two blocks and the cross entries. The product of their complex Thom classes is the complete-flag Thom class. Integration over the two internal flag varieties and then the Grassmannian evaluates on the same

complete-flag fundamental cycle. The identity  $p_n = p_d + p_e + de$  gives the total degree  $4p_n$ . All these degrees are even.

On coefficients, proper naturality of the support product commutes the internal  $\Phi(\gamma_d^{\text{fl}})$  and  $\Phi(\gamma_e^{\text{fl}})$  through the external support morphism. Before the internal pushforwards, all one-dimensional block potentials are zero. Their ordered support product is the constant coefficient on the complete-flag incidence. After the pushforwards, the target potentials remain the actual  $f_d$  and  $f_e$ . The composite therefore equals  $h_n$  when its inputs are  $h_d, h_e$ . The adjunction units compose to the complete-flag unit because they send the constant section to the same constant section on the refinement. Dividing by  $d!e!$  and using  $h_n = n!\sigma_n$  proves (1.5).

At a scalar atlas point, proper base change identifies the incoming stalk with hypercohomology on  $\text{Gr}(d, n)$  of the pulled-back coefficient family. Proposition 1.1.1 puts that family in degrees at least  $-2d - 2e = -2n$ . Theorem 1.2.1 identifies its cohomology sheaf in degree  $-2n$  with the constant sheaf generated by  $v_d \otimes v_e$ . The hypercohomology spectral sequence has no term below this degree. Its degree  $-2n$  term is  $H^0(\text{Gr}(d, n), \mathbb{Q}) = \mathbb{Q}$ . It has no incoming or outgoing differential, since the other relevant terms would have negative sheaf-cohomology degree or lower coefficient degree. Equation (1.5) proves the scalar assertion.  $\square$

In particular the first images involving  $f_3, f_4, f_5$  are

| $(d, e)$ | degree | incoming lowest class | image   |
|----------|--------|-----------------------|---------|
| $(2, 1)$ | $-6$   | $v_2 \otimes v_1$     | $3v_3$  |
| $(2, 2)$ | $-8$   | $v_2 \otimes v_2$     | $6v_4$  |
| $(3, 1)$ | $-8$   | $v_3 \otimes v_1$     | $4v_4$  |
| $(4, 1)$ | $-10$  | $v_4 \otimes v_1$     | $5v_5$  |
| $(2, 3)$ | $-10$  | $v_2 \otimes v_3$     | $10v_5$ |

Each incoming class denotes the constant section on the corresponding Grassmannian, with the coefficient morphisms just constructed.

#### 1.4 THREE CONSTITUENTS AND THE REMAINING SCALAR COEFFICIENTS

For an ordered triple  $(d, e, f)$  of positive ranks with sum  $n \leq 5$ , let  $\mathcal{F}_{d,e,f}$  be the full critical flag stack with these successive dimensions. Its maps  $a$  and  $b$  remember the constituents and the total object. The ordered triple support product and the flag Thom map define

$$M_{d,e,f} : Rb_* a^*(\mathcal{P}_d \boxtimes \mathcal{P}_e \boxtimes \mathcal{P}_f) \longrightarrow \mathcal{P}_n.$$

Both parenthesizations use this source, by proper base change. Their ambient Gysin degrees add to  $4(de + df + ef)$ , since

$$de + (d + e)f = ef + d(e + f).$$

Their normal entries partition the same forbidden entries, and their flag integrations evaluate on the same complex cycle. Ordered support associativity and its coefficient naturality therefore give equality of the two derived maps with  $M_{d,e,f}$ . This argument retains  $f_{d+e}$  and  $f_{e+f}$  on the intermediate representation spaces. It introduces no permutation of the inputs.

Refinement to complete flags gives the following restriction of this identity to the constructed coefficient morphisms:

$$M_{d,e,f}(\sigma_d, \sigma_e, \sigma_f) = \frac{n!}{d!e!f!} \sigma_n.$$

The left side includes the adjunction unit, as in (1.5). At a scalar point its source in degree  $-2n$  is again one-dimensional, by the same hypercohomology argument on the connected flag variety. For the first three scalar comparisons the two factorizations give

| $(d, e, f)$ | left factorization | right factorization | image   |
|-------------|--------------------|---------------------|---------|
| $(1, 1, 1)$ | $2 \cdot 3$        | $2 \cdot 3$         | $6v_3$  |
| $(2, 1, 1)$ | $3 \cdot 4$        | $2 \cdot 6$         | $12v_4$ |
| $(2, 2, 1)$ | $6 \cdot 5$        | $3 \cdot 10$        | $30v_5$ |

These are images of scalar coefficient classes. They follow from the complete-flag morphism and its nonzero scalar restriction.

The lowest coefficient has support on the entire scalar locus  $\mathbb{C}^3 \times BGL_n$  and on the distinct-eigenvalue locus. The translation and conjugation compatibilities specify the constant lowest cohomology sheaf on the scalar locus. The ambient derived coefficient remains in every incidence and refinement.

Higher scalar images require further calculations. For example, the rank-two scalar coefficient has degrees  $-4, -2, 1$ . On a scalar  $(2, 1)$  incidence, its degree  $-2$  coefficient tensored with the line coefficient starts in total degree  $-4$ . That degree also receives  $H^2(\mathbb{P}^2, \mathbb{Q})$  from the lowest coefficient. The morphism from these contributions to  $H^{-4}(P_{3,s})$  is not determined by (1.5). It requires the actual coefficient family, its hypercohomology differentials, and its map to the scalar target. Likewise, the odd rank-two class remains part of the  $(2, 2)$  and  $(2, 2, 1)$  inputs. Equation (1.1) describes the remaining scalar target groups through commuting-pair homology. It does not identify the Hall maps on those groups.

The constructed scalar images and derived flag identities are rational constructible statements through rank five. A comparison with a holomorphic field-theory carrier requires an additional morphism and compatibility with its operations. Tate structures, higher homotopies and further ranks require their own constructions.

### 1.5 THE FIRST HIGHER SCALAR HALL MAP

The scalar map for a plane and a line first has two incoming contributions in degree  $-4$ . Let  $s = (o, o, o) \in Z_3$ , let  $G = \text{Gr}(2, 3) = \mathbb{P}^2$ , and put

$$D_s = (Rb_{2,1*} a_{2,1}^*(\mathcal{P}_2 \boxtimes \mathcal{P}_1))_s.$$

Stalks use ordinary atlas pullback. Let  $\mathcal{K}$  be the pulled-back coefficient family on  $G$ , so proper base change gives  $D_s = R\Gamma(G, \mathcal{K})$ .

**PROPOSITION 1.5.1** (*The incoming degree and its proper comparison*). There is a canonical filtration of  $H^{-4}(D_s)$  with successive one-dimensional quotients

$$H^2(G, \mathcal{H}^{-6}\mathcal{K}), \quad H^0(G, \mathcal{H}^{-4}\mathcal{K}).$$

In particular  $\dim H^{-4}(D_s) = 2$ . Define smooth Milnor fibres

$$F = f_3^{-1}(1) \subset V_3, \quad I = \{(A, B, C, H) \in E_{2,1} : f_3(A, B, C) = 1\}, \quad q_1 : I \longrightarrow F.$$

The scalar coefficient map in degree  $-4$  is identified with the following composite:

$$H^{-4}(D_s) \xrightarrow{\sim} H^5(I; \mathbb{Q}) \xrightarrow{q_{1*}} H^{13}(F; \mathbb{Q}) \xrightarrow{\sim} H_{22}^{\text{BM}}(C_3; \mathbb{Q}). \quad (1.6)$$

Here  $q_{1*}$  is the proper cohomological Gysin map with complex orientations. The right isomorphism is the scalar support comparison of Proposition 1.1.1.

*Proof.* The rank-two scalar coefficient has degrees  $-4, -2, 1$ , each of rank one. The line coefficient has degree  $-2$ . Hence the cohomology sheaves of  $\mathcal{K}$  have degrees  $-6, -4, -1$  and rank one. They are locally constant: local frames trivialize the tautological plane and quotient, and changes of frames preserve the scalar point. Every local system on  $\mathbb{P}^2$  is constant, since  $\mathbb{P}^2$  is simply connected.

In the hypercohomology spectral sequence  $E_2^{p,q} = H^p(G, \mathcal{H}^q \mathcal{K})$ , the terms of total degree  $-4$  are  $(p, q) = (2, -6)$  and  $(0, -4)$ . The first has no outgoing differential because no coefficient occurs below  $-6$ . Its only possible incoming differential has source  $(0, -5)$ , which is zero. The second has no incoming differential. Its outgoing differential of length two has coefficient degree  $-5$ . The length-three target is  $(3, -6)$ , where  $H^3(\mathbb{P}^2) = 0$ . Longer targets have coefficient degree below  $-6$ . Thus both terms survive, giving the filtration and dimension. The filtration does not select a splitting of the two contributions.

Write  $E = E_{2,1}$  and  $g = f_3|_E$ . The variety  $E$  is a vector bundle of rank  $21$  over  $G$ , and  $\dim_{\mathbb{C}} E = 23$ . Its block expression is  $g = f_2 + 0$ . At potential value one the differential of the rank-two potential is nonzero, so  $I$  is smooth of complex dimension  $22$ . Likewise  $F$  is smooth of complex dimension  $26$ . The restriction  $q_1$  is proper, as a base change of the proper incidence projection.

Positive scaling of the matrix entries contracts  $E$  to its zero section  $G$  while preserving the sign of  $\operatorname{Re} g$ . The same radial pair argument as in Proposition 1.1.1 computes the scalar source by

$$D_s \simeq \operatorname{Fib}(R\Gamma(E, \mathbb{Q}) \longrightarrow R\Gamma(I, \mathbb{Q})) [10].$$

The equality uses the rank-one zero-potential comparison, so it retains the actual incoming vanishing-cycle coefficient. Since  $E$  retracts to  $\mathbb{P}^2$ , its cohomology in degrees five and six vanishes. The relative-cochain exact sequence identifies  $H^{-4}(D_s)$  with  $H^5(I)$ . The target is  $\operatorname{Fib}(\mathbb{Q} \rightarrow R\Gamma(F, \mathbb{Q})) [18]$ . Its degree  $-4$  group is  $H^{13}(F)$ .

The ambient incidence Gysin map has degree eight. Restriction to the smooth potential fibre is the proper Gysin map  $q_{1*}$ , also of degree eight, since  $26 - 22 = 4$ . The normal Thom construction and flag integration commute with restriction of the potential. Therefore they induce a morphism between the two relative-cochain sequences just used. Both boundary identifications use the same normalized fibre convention. The shift eight is even, so the square has no additional sign. This proves the first two arrows of (1.6). For  $n = 3$  and  $j = -4$ , the Borel–Moore index in (1.1) is  $18 - (-4) = 22$ , proving the last arrow.  $\square$

The projection of  $I$  to its preserving pair  $(A, C, H)$  has affine fibres over the locus where the diagonal rank-two pair does not commute. Forgetting  $H$  from this pair locus is not proper. Fix  $H = \langle e_1, e_2 \rangle$  and set

$$A = \operatorname{diag}(0, 1, 2), \quad C_t = te_{12} + e_{13}, \quad B_t = t^{-1}e_{21} \quad (t \neq 0).$$

These matrices preserve  $H$  and satisfy

$$[C_t, A] = te_{12} + 2e_{13}, \quad \operatorname{Tr}(B_t[C_t, A]) = 1.$$

The pair  $(A, C_t)$  has a limit in  $U_3$  at  $t = 0$ , since  $[C_0, A] = 2e_{13} \neq 0$ . Its two diagonal blocks commute at that limit. Thus the limit lies outside the projected incidence locus. The flag remains fixed, so no other limiting flag repairs this missing point. The matrix  $B_t$  diverges, consistently with the properness of  $q_1$  before this affine projection.

The image of the two-dimensional group in (1.6) therefore requires the proper Milnor-fibre Gysin map, or a proved replacement retaining its support data. The filtration specifies the two incoming contributions. Their images and the rank of  $q_{1*}$  in this degree remain to be determined.

### 1.6 EVALUATION OF THE DEGREE-MINUS-FOUR IMAGE

The first higher scalar map has rank one. Its kernel has a geometric generator obtained by decorating the internal complete flag. The calculation begins with the rank-two coefficient carried by that decoration.

Let  $\nu_2$  be the complete-flag coefficient map. On its scalar fibre  $\mathbb{P}^1$ , let  $h = c_1(\mathcal{O}_{\mathbb{P}^1}(1))$ , with  $\int_{\mathbb{P}^1} h = 1$ . Define

$$w_2 = H^{-2}(\nu_2)(h) \in H^{-2}(P_{2,0}).$$

The input shift of  $\nu_2$  is  $[4]$ , so this notation uses degree two on the flag fibre.

LEMMA 1.6.1 (*The decorated rank-two flag*). The class  $w_2$  is nonzero. Thus it generates  $H^{-2}(P_{2,0}) = \mathbb{Q}$ .

*Proof.* In traceless matrix coordinates the potential is  $2 \det T$  on  $W = \text{Mat}_3(\mathbb{C})$ . Scalar matrix coordinates are free. Write  $D \subset W$  for the determinant hypersurface. Let  $G_0 = \text{Gr}(2, 3) = \mathbb{P}^2$ , with tautological plane bundle  $U$ , and let

$$\tilde{D} = \text{Hom}(\mathbb{C}^3, U) \longrightarrow D$$

be the proper incidence remembering a plane containing the columns of  $T$ . Its total space is smooth of complex dimension eight. Write  $z = c_1(\mathcal{O}_{G_0}(1))$ . The tautological exact sequence gives  $c_1(U) = -z$  and  $c_2(U) = z^2$ .

Let  $D' \subset D$  be the matrices of rank at most one and  $E' \subset \tilde{D}$  its inverse image. Their complex dimensions are five and six. Over  $D \setminus D'$ , the incidence is an isomorphism. The proper resolution of  $E'$  that remembers the image line has total space

$$Y = \text{Hom}(\mathbb{C}^3, L) \longrightarrow \text{Fl}(1, 2; 3),$$

where  $L \subset U$  is the tautological line. It is a rank-three vector bundle over the complete flag variety. Its map to  $E'$  is an isomorphism away from zero matrices. The exceptional locus has complex dimension three, and its image has complex dimension two. Closed-open exact sequences therefore identify  $H_{11}^{\text{BM}}(E')$  with  $H_{11}^{\text{BM}}(Y) = 0$ . The last equality follows from the even cohomology of the complete flag variety and the vector-bundle Thom isomorphism.

The top class of  $E'$  maps to zero in  $H_{12}^{\text{BM}}(\tilde{D})$ . To compute it, let  $p : \text{Fl}(1, 2; 3) \rightarrow G_0$  forget the line and put  $q = c_1(U/L)$ . The inclusion of  $Y$  into the pullback of  $\tilde{D}$  has quotient normal bundle  $\text{Hom}(\mathbb{C}^3, U/L)$ . Its Euler class is  $q^3$ . Hence the class of  $E'$  in the smooth space  $\tilde{D}$  is Poincaré dual to

$$p_*(q^3) = c_1(U)^2 - c_2(U) = z^2 - z^2 = 0.$$

Indeed,  $q^2 - c_1(U)q + c_2(U) = 0$  by the tautological exact sequence, while integration on the projective-line fibre gives  $p_*q = 1$  and  $p_*1 = 0$ . The resolution  $Y \rightarrow E'$  is generically one-to-one, so it pushes its fundamental class to that of  $E'$ .

The closed-open sequence for  $E' \subset \tilde{D}$  now shows that  $H_{12}^{\text{BM}}(\tilde{D}) \rightarrow H_{12}^{\text{BM}}(\tilde{D} \setminus E')$  is an isomorphism. Its kernel is the zero exceptional class and its cokernel maps to  $H_{11}^{\text{BM}}(E') = 0$ . Since  $\dim_{\mathbb{C}} D' = 5$ ,

the corresponding map from  $H_{12}^{\text{BM}}(D)$  to the open stratum is also an isomorphism. Consequently the proper pushforward is an isomorphism

$$H_{12}^{\text{BM}}(\tilde{D}) = \mathbb{Q} \xrightarrow{\sim} H_{12}^{\text{BM}}(D) = \mathbb{Q}.$$

A line in  $\mathbb{C}^2$  determines its traceless stabilizer plane in  $\mathfrak{sl}_2$ . Under the trace pairing, the perpendicular line consists of nilpotent matrices with that image and kernel. Thus the map from the original flag  $\mathbb{P}^1$  to  $G_0$  is the smooth conic  $j : \mathbb{P}^1 \hookrightarrow \mathbb{P}^2$ . Complex-oriented integration gives

$$j_*1 = 2z, \quad j_*h = z^2.$$

The complete-flag incidence is the restriction of  $\tilde{D}$  to this conic. Transitivity of complex Thom maps and proper integration identifies its Gysin map with  $j_*$  followed by the universal determinant incidence map.

The support form of this universal map has the following description. Poincaré duality on  $\tilde{D}$  takes  $z^2$  to its nonzero degree-twelve Borel–Moore class. Proper pushforward takes this to the nonzero degree-twelve class of  $D$ . The closed-open sequence in  $\mathcal{W}$ , of real dimension eighteen, takes it to the nonzero class in  $H^5(\text{GL}_3(\mathbb{C}))$ . The support inclusion  $D \subset \{\text{Re}(2 \det T) \leq 0\}$  restricts this class to the determinant Milnor fibre. These are the maps of the same relative-cochain sequences defining the normalized coefficient map, so the description also fixes its sign.

The determinant fibration has a product trivialization  $\text{GL}_3(\mathbb{C}) \simeq \text{SL}_3(\mathbb{C}) \times \mathbb{C}^*$ . Since  $H^4(\text{SL}_3) = 0$ , restriction induces an isomorphism on  $H^5$ . Multiplication by the positive constant two in the potential preserves the chosen support half-plane. Therefore the image of  $j_*h = z^2$  is nonzero in reduced Milnor degree five, hence in normalized scalar degree  $-2$ . This image is exactly  $w_2$ .  $\square$

The rank-three target in the required degree also has dimension one. The two possible eigenvalue collisions determine this dimension.

**LEMMA 1.6.2** (*The next Borel–Moore group of the commuting-pair variety*). For the variety  $C_3$  of commuting *pairs* of three-by-three matrices,

$$H_{22}^{\text{BM}}(C_3; \mathbb{Q}) = \mathbb{Q}, \quad H^{-4}(P_{3,0}) = \mathbb{Q}.$$

*Proof.* Remove the locus where  $A$  has at most one eigenvalue. The Jordan dimension calculation gives complex dimension at most ten for this locus. Its removal does not change Borel–Moore homology in degrees twenty-two and twenty-three. Denote the remaining space by  $X$ , and let  $O \subset X$  be the open set where  $A$  has three distinct eigenvalues.

The smooth variety  $O$  has complex dimension twelve. Up to an equivariantly contractible affine factor, its ordered eigenvalue cover is

$$\text{GL}_3(\mathbb{C})/T \times \text{Conf}_3(\mathbb{C}),$$

where  $T$  is the diagonal torus and  $\text{Conf}_3$  is the space of ordered distinct triples. The symmetric group permutes both factors, freely on the second. Rational cohomology of  $O$  is the invariant part of this cover. The first factor has the cohomology of the complete flag variety. Its  $H^1$  is zero, and its  $H^2$  is generated by three Chern roots with their sum zero. The permutation action has no invariant in this degree.

For the second factor, write

$$(z_1, z_2, z_3) = (a, a + b, a + bc), \quad a \in \mathbb{C}, \quad b \in \mathbb{C}^*, \quad c \in \mathbb{C} \setminus \{0, 1\}.$$

These are global coordinates. Its degree-two cohomology has basis

$$\alpha = d \log b \wedge d \log c, \quad \beta = d \log b \wedge d \log(c - 1),$$

where each logarithmic degree-one class includes the factor  $(2\pi i)^{-1}$ . The transposition (12) exchanges  $\alpha$  and  $\beta$ . The transposition (23) sends  $\alpha \mapsto -\alpha$  and  $\beta \mapsto \beta - \alpha$ . The term  $d \log c \wedge d \log(c - 1)$  has zero cohomology class because it comes from the punctured  $c$ -line. These two transformations have no common fixed vector. In degree one, the three classes  $d \log(z_i - z_j)$  are permuted, so their invariant subspace has dimension one. It follows that

$$H^1(O; \mathbb{Q}) = \mathbb{Q}, \quad H^2(O; \mathbb{Q}) = 0.$$

Poincaré duality gives  $H_{23}^{\text{BM}}(O) = \mathbb{Q}$  and  $H_{22}^{\text{BM}}(O) = 0$ .

The closed complement  $D_0 = X \setminus O$  has two irreducible components of complex dimension eleven. On the first, the repeated eigenvalue of  $A$  has a Jordan block of size two. On the second, its two-dimensional eigenspace is semisimple. Both strata are irreducible: the centralizer vector bundle over each Jordan stratum gives their parameterization. Their closures are distinct. The generic commuting matrix in the semisimple stratum has two different eigenvalues on that eigenspace. On the Jordan stratum the corresponding restriction is a polynomial in the Jordan block, so its two eigenvalues coincide even in a limit. The opposite containment is excluded by the smaller minimal polynomial on the semisimple stratum. Thus  $H_{22}^{\text{BM}}(D_0) = \mathbb{Q}^2$ .

The boundary map from  $H_{23}^{\text{BM}}(O)$  to these two fundamental classes is nonzero. More precisely, the discriminant of the characteristic polynomial has multiplicities one and two on the two components. At a generic Jordan point,  $A$  is cyclic, so the commuting pair variety is locally a centralizer vector bundle over matrix space. The family

$$A_t = \begin{pmatrix} 0 & 1 & 0 \\ t & 0 & 0 \\ 0 & 0 & 2 \end{pmatrix}$$

has characteristic discriminant  $4t(4 - t)^2$ , of order one. At a generic semisimple point, choose the commuting matrix  $C$  with three distinct eigenvalues. The local commuting-pair chart then diagonalizes  $A$  in its eigenbasis. The family

$$A_t = \text{diag}(t, -t, 2)$$

has discriminant  $4t^2(4 - t^2)^2$ , of order two. Both are transverse local coordinates in the specified smooth charts. The linking class  $d \log \text{disc}(A)$  therefore has boundary a nonzero common multiple of (1, 2), with complex orientations. The closed-open sequence gives

$$H_{22}^{\text{BM}}(X) = \mathbb{Q}^2 / \mathbb{Q}(1, 2) = \mathbb{Q}.$$

Restoring the removed locus preserves this group. Proposition I.I.1 identifies it with  $H^{-4}(P_{3,0})$ .  $\square$

The complete flag in dimension three has successive line bundles  $L_1, L_2, L_3$ . Let  $\ell_i = c_1(L_i)$  and define the three degree-zero morphisms

$$\kappa_i : \mathbb{Q}_{\mathcal{M}_3}[4] \longrightarrow Rb_{3*} \mathbb{Q}_{\mathcal{T}_3}[4] \xrightarrow{\ell_i} Rb_{3*} \mathbb{Q}_{\mathcal{T}_3}[6] \xrightarrow{\nu_3} \mathcal{P}_3.$$

The first map is the adjunction unit. Cup product by a first Chern class has cohomological degree two.

LEMMA 1.6.3 (*Three line decorations*). At the scalar atlas point,

$$\kappa_3(1) \neq 0, \quad \kappa_2(1) = 0 \quad \text{in } H^{-4}(P_{3,0}).$$

At the critical point  $x = (\text{diag}(0, 0, 1), 0, 0)$ , identify the degree  $-4$  coefficient with the rank-two class  $w_2$  tensored with the normalized line class. Then

$$\kappa_1(1) = -2w_2, \quad \kappa_2(1) = 0, \quad \kappa_3(1) = 2w_2.$$

*Proof.* The spectral projector for the two separated eigenvalue clusters gives a local analytic decomposition into a plane and a line. In this chart,  $A$  is block diagonal. Its cross-block commutator operator is invertible. The expression  $\text{Tr}(B[C, A])$  therefore splits exactly into the rank-two potential and a non-degenerate bilinear form in the cross-block  $B, C$  coordinates. Rescaling the cross-block  $C$  coordinates gives two eigenline-pair quadratics. The rescaling is complex linear, so it preserves orientation. Their normalized coefficient, including the free line variables, identifies the target coefficient with  $\mathcal{P}_2 \boxtimes \mathcal{P}_1$ .

An invariant complete flag separates according to the position of the isolated eigenline. Its fibre at  $x$  has three components, each a copy of  $\mathbb{P}^1$ . On each component, the remaining flag is the complete flag in the double eigenspace. The two cross eigenline pairs contribute  $(-1)^2 = 1$  in the  $B$  polarization. The conversion to the specified generators also has ratio  $(-1)^{p_2+p_1-p_3} = 1$ . The flag tangents and lower  $A$  normals cancel complex linearly. Thus each component contributes the actual map  $\nu_2$ , with coefficient  $+1$ , to the target identification just given.

Let  $h$  be the positive hyperplane class on that component. The tautological subline in the double eigenspace has Chern class  $-h$ , and its quotient line has Chern class  $h$ . The isolated line is constant. Accordingly the three restrictions are

| position of isolated line | $\ell_1$ | $\ell_2$ | $\ell_3$ |
|---------------------------|----------|----------|----------|
| 1                         | 0        | $-h$     | $h$      |
| 2                         | $-h$     | 0        | $h$      |
| 3                         | $-h$     | $h$      | 0.       |

Applying  $\nu_2(h) = w_2$  and adding the three components proves the values at  $x$ .

Each  $\kappa_i$  induces a section of  $\mathcal{H}^{-4}(\mathcal{P}_3)$  on the ordinary atlas. These sections are equivariant under positive scaling of the matrices. Since  $\kappa_3$  is nonzero at  $x$ , its origin stalk is nonzero: a section with zero origin germ vanishes near the origin, and scaling would then make it vanish everywhere. Lemma 1.6.2 makes that scalar group one-dimensional. Write  $\kappa_2(1) = c\kappa_3(1)$  at the origin. The difference section has zero origin germ, hence vanishes everywhere by the same argument. Its value at  $x$  is  $-2cw_2$ . Lemma 1.6.1 forces  $c = 0$ .  $\square$

The incoming filtration admits a specified geometric lift. Define  $\tau_2 : \mathbb{Q}_{\mathfrak{M}_2}[2] \rightarrow \mathcal{P}_2$  by the complete-flag construction with the first Chern class of its quotient line inserted. Its scalar degree  $-2$  class is  $w_2$ . On the scalar Grassmannian  $G = \text{Gr}(2, 3)$ , let  $H = c_1(\mathbb{C}^3/U)$ , with  $\int_G H^2 = 1$ . Define two classes in  $H^{-4}(D_s)$  by

$$a = H(v_2 \otimes v_1), \quad b = (Rb_{2,1*} a_{2,1}^*(\tau_2 \boxtimes \sigma_1) \circ \eta_{2,1}[4])_s(1).$$

The first class lies in the degree-two flag contribution. The second is an actual derived coefficient class, not a chosen splitting of the cohomology family.

**THEOREM 1.6.4** (*The degree-minus-four scalar image*). The classes  $a, b$  form a basis of the two-dimensional scalar source  $H^{-4}(D_s)$ . Put  $w_3 = \frac{1}{2}\kappa_3(1)$  at the scalar point. Then  $w_3$  generates  $H^{-4}(P_{3,o})$ , and the actual Hall map is

$$H^{-4}(\mu_{2,1})(a) = w_3, \quad H^{-4}(\mu_{2,1})(b) = 0.$$

Thus the degree- $-4$  image is the whole target and its kernel is  $\mathbb{Q}b$ . Under the proper Milnor-fibre identification (1.6), the Gysin map has rank one.

*Proof.* The first class generates the first term of the filtration in Proposition 1.5.1. The edge image of  $b$  in its second quotient is  $w_2 \otimes v_1$ , because the scalar stalk of  $\tau_2$  is  $w_2$  at every plane. This is nonzero by Lemma 1.6.1. Hence  $a, b$  form a basis.

Refine the plane by its complete flag. Inserting its quotient-line Chern class gives  $\ell_2$  on the complete flag of dimension three. The proper refinement and support naturality used in Theorem 1.3.1 commute with cup product by this pulled-back Chern class. Consequently  $\mu_{2,1}(b) = \kappa_2(1)$ . Inserting the outer quotient-line class instead gives  $\ell_3$ . The normalized plane input is  $\sigma_2 = h_2/2$ . Thus  $\mu_{2,1}(a) = \kappa_3(1)/2 = w_3$ . Lemma 1.6.3 proves the displayed values and the nonvanishing of  $w_3$ . The target dimension follows from Lemma 1.6.2.  $\square$

The normalization of  $w_3$  is geometric. The section  $\kappa_3/2$  specializes to  $w_2$  when a scalar plane is separated from a line. The class  $w_2$  is the positive flag decoration followed by the determinant-incidence support map. This fixes the generator without identifying it with an unspecified Borel–Moore basis.

The computation determines one higher scalar degree for the rank- $(2, 1)$  Hall map. The remaining degrees and the higher scalar images for ranks four and five retain their separate support and extension calculations.

## 1.7 THE FIRST HIGHER IMAGES THROUGH RANK FIVE

The first higher scalar coefficient for each of  $f_4$  and  $f_5$  is one-dimensional. Complete-flag line classes determine its binary and three-constituent Hall images. All ranks in this section satisfy  $2 \leq n \leq 5$ . The scalar points, coefficients and orientations remain those of Sections 1.1–1.6.

**LEMMA 1.7.1** (*Two adjacent scalar degrees*). At a scalar atlas point,

$$H^{-2n+1}(P_{n,o}) = 0, \quad H^{-2n+2}(P_{n,o}) = \mathbb{Q}.$$

*Proof.* Set  $N = n^2 + n = \dim_{\mathbb{C}} C_n$ . Remove the pairs for which  $\mathcal{A}$  has at most  $n - 2$  distinct eigenvalues. Their complex dimension is at most  $N - 2$ , by the Jordan calculation. This removal preserves Borel–Moore homology in degrees  $2N - 1$  and  $2N - 2$ . For  $n = 2$ , the removed locus is empty. Let  $O$  be the distinct-eigenvalue locus in the remaining space  $X$ . Its ordered cover has, up to a contractible affine factor, the form

$$\mathrm{GL}_n(\mathbb{C})/T_n \times \mathrm{Conf}_n(\mathbb{C}).$$

Here  $T_n$  is the diagonal torus. The first factor has  $H^1 = 0$ . Its  $H^2$  consists of the  $n$  Chern roots modulo their sum. Hence its symmetric-group invariants in degree two vanish.

The degree-one cohomology of ordered configurations has basis  $a_{ij} = (2\pi i)^{-1} d \log(z_i - z_j)$ , for  $i < j$ , and its cohomology in degree two is generated by their products. Here is a direct induction for the required generation statement. Forgetting the last point is a locally trivial bundle with fibre  $\mathbb{C} \setminus \{z_1, \dots, z_{n-1}\}$ . Local trivializations follow by moving the points in disjoint disks and extending

those motions to the plane with compact support. The classes  $a_{1n}, \dots, a_{n-1,n}$  restrict to the basis dual to the small positive circles in the punctured-plane fibre. These global classes supply permanent cycles for the whole fibre cohomology. The Serre spectral sequence therefore collapses. The cohomology is a free module over the base cohomology on  $1, a_{1n}, \dots, a_{n-1,n}$ . Induction from  $n = 1$  proves the assertion, including the stated degree-one basis.

The symmetric group permutes the unordered-edge basis  $a_{ij}$ . It has a one-dimensional invariant subspace in degree one. The exterior square of the full degree-one representation has no invariants. Indeed, for two distinct edges sharing a vertex, the transposition of the other endpoints interchanges the two edges. For disjoint edges, interchanging their two pairs of endpoints has the same effect. Each operation negates the corresponding wedge. Its average over the symmetric group is therefore zero. Since invariants over  $\mathbb{Q}$  preserve surjections, the quotient  $H^2(\text{Conf}_n)$  also has no invariants. Consequently  $H^1(O) = \mathbb{Q}$  and  $H^2(O) = 0$ .

The complement  $X \setminus O$  has two components of complex dimension  $N - 1$ . They have one double eigenvalue of  $A$ , with respectively a Jordan block and a semisimple double eigenspace. All remaining eigenvalues are distinct. Centralizer bundles show that both strata are irreducible. Their closures are distinct by the same commuting-eigenvalue and minimal-polynomial conditions used in Lemma 1.6.2. The discriminant has multiplicities one and two on these components. To check this, use the two-by-two blocks

$$\begin{pmatrix} 0 & 1 \\ t & 0 \end{pmatrix}, \quad \text{diag}(t, -t),$$

and append fixed, pairwise distinct eigenvalues away from zero. All additional factors in the discriminant are units near  $t = 0$ . The cyclic- $A$  chart in the first case and cyclic- $C$  chart in the second give smooth transverse coordinates, as before.

The closed-open sequence thus has the exact segment

$$0 \longrightarrow H_{2N-1}^{\text{BM}}(X) \longrightarrow \mathbb{Q} \xrightarrow{(1,2)} \mathbb{Q}^2 \longrightarrow H_{2N-2}^{\text{BM}}(X) \longrightarrow 0.$$

The first group is zero and the last is one-dimensional. Restoring the removed locus preserves these groups. The scalar support isomorphism (1.1) gives the two asserted degrees.  $\square$

For the complete flag in dimension  $n$ , let  $\kappa_{n,k} : \mathbb{Q}\mathfrak{M}_n[2n-2] \rightarrow \mathcal{P}_n$  be the complete-flag morphism with  $c_1(L_k)$  inserted. Here  $L_k$  is its  $k$ th successive quotient line. Thus  $\kappa_{3,k} = \kappa_k$  and  $\kappa_{2,2} = \tau_2$ .

**PROPOSITION 1.7.2** (*Normalized first higher classes*). The morphism

$$\tau_n = \frac{\kappa_{n,n}}{(n-1)!} : \mathbb{Q}\mathfrak{M}_n[2n-2] \longrightarrow \mathcal{P}_n$$

has a nonzero scalar class  $w_n$  in degree  $-2n + 2$ . It generates the group of Lemma 1.7.1. For each  $1 \leq k \leq n$ , its scalar line decorations satisfy

$$\kappa_{n,k}(1) = (2k - n - 1)(n - 2)! w_n.$$

These generators agree with  $w_2$  and  $w_3$  already constructed.

*Proof.* Take a critical triple with one scalar plane and  $n - 2$  separated scalar lines. The spectral projectors split its local coefficient into the rank-two coefficient and the line coefficients, together with the nondegenerate cross-block normal quadratic. The same invertible commutator operator and complex rescaling

as in Lemma 1.6.3 give this decomposition. There are  $p_n - 1$  cross eigenline pairs. Their  $B$ -polarized sign  $(-1)^{p_n-1}$  is canceled by the generator-conversion sign  $(-1)^{1-p_n}$ . Thus every flag component contributes  $v_2$  with coefficient  $+1$ .

Choose the two positions of the lines in the double eigenspace. The other  $n - 2$  labeled lines can occupy the remaining positions in  $(n - 2)!$  orders. Each such choice gives one  $\mathbb{P}^1$  component of the flag fibre. On it,  $c_1(L_k)$  is  $-h$  if position  $k$  is the first double-eigenspace line,  $h$  if it is the second, and zero otherwise. There are  $(n - k)(n - 2)!$  negative contributions and  $(k - 1)(n - 2)!$  positive contributions. The value of  $\kappa_{n,k}$  is therefore  $(2k - n - 1)(n - 2)! w_2$  on this stratum, including the line factors.

For  $k = n$  this is  $(n - 1)! w_2 \neq 0$ . Positive-scaling equivariance of the actual cohomology section forces its scalar origin germ to be nonzero. The scalar group is one-dimensional by Lemma 1.7.1. Subtracting the claimed multiple of  $\tau_n$  from each decoration gives a section whose scalar germ is some multiple of  $w_n$ . Its value on the separated-plane stratum determines this multiple, by the same zero-germ scaling argument. This proves the formula and the normalizations.  $\square$

Fix  $d + e = n \leq 5$  and let  $D_{d,e}$  be the actual scalar incoming complex for  $\mu_{d,e}$ . Let  $U \subset \mathbb{C}^n$  be the tautological  $d$ -plane on  $\text{Gr}(d, n)$  and put  $H = c_1(\mathbb{C}^n/U)$ . Define the following scalar classes in degree  $-2n + 2$ :

$$\begin{aligned} A_{d,e} &= H(v_d \otimes v_e), \\ B_{d,e} &= (Rb_* a^*(\tau_d \boxtimes \sigma_e) \circ \eta[2n - 2])_o(1) \quad (d \geq 2), \\ C_{d,e} &= (Rb_* a^*(\sigma_d \boxtimes \tau_e) \circ \eta[2n - 2])_o(1) \quad (e \geq 2). \end{aligned}$$

The maps  $a, b, \eta$  refer to the same full critical correspondence and its adjunction unit as in (1.5). The letters  $A_{d,e}, B_{d,e}, C_{d,e}$  denote coefficient classes, not representation matrices.

**THEOREM 1.7.3 (Binary first higher images).** The displayed classes form a basis of  $H^{-2n+2}(D_{d,e})$ , whose dimension is  $1 + \mathbf{1}_{d \geq 2} + \mathbf{1}_{e \geq 2}$ . The actual Hall map has values

$$\begin{aligned} \mu_{d,e}(A_{d,e}) &= \frac{(n - 2)!}{(d - 1)!(e - 1)!} w_n, \\ \mu_{d,e}(B_{d,e}) &= \frac{(d - e - 1)(n - 2)!}{(d - 1)!e!} w_n \quad (d \geq 2), \\ \mu_{d,e}(C_{d,e}) &= \frac{(n - 1)!}{d!(e - 1)!} w_n \quad (e \geq 2). \end{aligned}$$

In particular this map is surjective and has rank one.

*Proof.* The incoming coefficient has no group one degree above its lowest degree, by Lemma 1.7.1 and  $P_1 = \mathbb{Q}[2]$ . Its groups two degrees above the lowest come from the first higher class of either nontrivial-rank factor. The Grassmannian has no odd cohomology and has one-dimensional  $H^2$ . Thus the only terms of total degree  $-2n + 2$  are its  $H^2$  with the lowest coefficient and its  $H^0$  with these higher coefficients. The same differential argument as in Proposition 1.5.1 makes all these terms permanent. The classes  $B_{d,e}, C_{d,e}$  lift their separate nonzero edge classes, whereas  $A_{d,e}$  generates the flag contribution. This proves the basis.

Refine the partial flag to a complete flag and commute Chern insertions through the proper support comparison. The three incoming classes become respectively

$$\frac{\sum_{k=d+1}^n c_1(L_k)}{d!e!}, \quad \frac{c_1(L_d)}{(d - 1)!e!}, \quad \frac{c_1(L_n)}{d!(e - 1)!}$$

under the complete-flag coefficient map. Proposition 1.7.2 evaluates them. For the first numerator,  $\sum_{k=d+1}^n (2k - n - 1) = de$ . The other two numerators are  $d - e - 1$  and  $n - 1$ . These identities give the stated coefficients. The first coefficient is a positive integer, proving surjectivity.  $\square$

For the three target ranks the resulting matrices, in the displayed geometric bases and the normalized  $w_n$ , are

| $(d, e)$ | degree | matrix on $(A, B, C)$ with absent terms omitted |
|----------|--------|-------------------------------------------------|
| $(2, 1)$ | $-4$   | $(1, 0)$                                        |
| $(2, 2)$ | $-6$   | $(2, -1, 3)$                                    |
| $(3, 1)$ | $-6$   | $(1, 1)$                                        |
| $(4, 1)$ | $-8$   | $(1, 2)$                                        |
| $(2, 3)$ | $-8$   | $(3, -2, 6)$ .                                  |

The same Chern insertions determine the three-constituent comparison. For a scalar flag with dimensions  $(d, e, f)$ , set  $X = c_1(\mathbb{C}^n/E_d)$  and  $Y = c_1(\mathbb{C}^n/E_{d+e})$ . Their products with the lowest coefficient give the two flag classes. Include the actual lifts with  $\tau$  in each constituent of rank at least two. For  $(1, 1, 1)$ ,  $(2, 1, 1)$ ,  $(2, 2, 1)$ , these form bases by the preceding spectral-sequence argument on the partial flag variety. Its Schubert cells give two independent degree-two classes  $X, Y$  and no odd cohomology. The common derived flag map has matrices

| $(d, e, f)$ | degree | basis order                                           | image coefficients in $w_n$ |
|-------------|--------|-------------------------------------------------------|-----------------------------|
| $(1, 1, 1)$ | $-4$   | $(X, Y)$                                              | $(2, 2)$                    |
| $(2, 1, 1)$ | $-6$   | $(X, Y, \tau_2 \text{ in the first factor})$          | $(4, 3, -2)$                |
| $(2, 2, 1)$ | $-8$   | $(X, Y, \tau_2 \text{ in the first, second factors})$ | $(9, 6, -6, 6)$ .           |

Indeed, complete-flag refinement divides the two flag insertions by  $d!e!f!$  and replaces them by the two corresponding suffix sums of the  $c_1(L_k)$ . A constituent insertion divides instead by the factorial one smaller in that constituent and selects its last line. The values follow from Proposition 1.7.2. The two parenthesisations give these same maps because the ordered support products, Chern pullbacks and complex Thom integrations commute. This determines the first higher scalar images and their stated three-constituent compatibility through rank five.

## BIBLIOGRAPHY

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